# General Dessert #9: Station Configuration and Transmitter Controls

**Receiver tools (notch, noise blanker, noise reduction, attenuator, reverse sideband), antenna tuners, vacuum-tube amp TUNE/LOAD, ALC, electronic keyers, dual VFO.**

**Group G4A · 13 pool questions · Self-study reference · Practice exams at eham.net/exams**

## Key Facts Reference

### Receiver Controls

| Control | What it does | Q |
|---|---|---|
| Notch filter | **Reduces interference from carriers (single tones) in the receiver passband.** | G4A01 |
| Reverse sideband on CW | **May reduce or eliminate interference from other signals** by putting the QRM on the opposite audio side of the BFO. | G4A02 |
| Noise blanker | **Reduces receiver gain during a noise pulse** (gates the audio briefly while the impulse is present). | G4A03 |
| Noise reduction (turned up too far) | **Received signals may become distorted.** | G4A07 |
| Receive attenuator | **Prevents receiver overload from strong incoming signals.** | G4A13 |

### Transmitter and Amplifier Controls

| Control | Setting / purpose | Q |
|---|---|---|
| Vacuum-tube amp TUNE control | Adjust for a **pronounced dip** in plate current at resonance. | G4A04 |
| Vacuum-tube amp LOAD / COUPLING | Adjust for **desired power output without exceeding maximum allowable plate current.** | G4A08 |
| Automatic Level Control (ALC) purpose | **Prevents excessive drive** into the RF power amplifier. | G4A05 |
| ALC during AFSK digital modes | Should be **inactive** – **ALC action distorts the signal** on AFSK. | G4A11 |
| Keying-line-to-RF delay on external amp | **Allows the amp time to switch the antenna** (T/R relay) between transceiver bypass and amplifier output. | G4A09 |
| Electronic keyer function | **Automatic generation of dots and dashes for CW operation.** | G4A10 |
| Dual-VFO common use | **Transmit on one frequency and listen on another** (split operation, DX pileups). | G4A12 |

### Antenna Tuner

| Question | Answer | Q |
|---|---|---|
| Purpose of an antenna tuner | **Increase power transfer from the transmitter to the feed line** by matching the transmitter to the load it sees. Note: tuner does NOT reduce SWR on the feed line between tuner and antenna. | G4A06 |

# Dessert 9 Question Bank (G4A)

**G4A01.** What is the purpose of the notch filter found on many HF transceivers?
A. To restrict the transmitter voice bandwidth
B. To reduce interference from carriers in the receiver passband
C. To eliminate receiver interference from impulse noise sources
D. To remove interfering splatter generated by signals on adjacent frequencies

**G4A02.** What is the benefit of using the opposite or “reverse” sideband when receiving CW?
A. Interference from impulse noise will be eliminated
B. More stations can be accommodated within a given signal passband
C. It may be possible to reduce or eliminate interference from other signals
D. Accidental out-of-band operation can be prevented

**G4A03.** How does a noise blanker work?
A. By temporarily increasing received bandwidth
B. By redirecting noise pulses into a filter capacitor
C. By reducing receiver gain during a noise pulse
D. By clipping noise peaks

**G4A04.** What is the effect on plate current of the correct setting of a vacuum-tube RF power amplifier’s TUNE control?
A. A pronounced peak
B. A pronounced dip
C. No change will be observed
D. A slow, rhythmic oscillation

**G4A05.** Why is automatic level control (ALC) used with an RF power amplifier?
A. To balance the transmitter audio frequency response
B. To reduce harmonic radiation
C. To prevent excessive drive
D. To increase overall efficiency

**G4A06.** What is the purpose of an antenna tuner?
A. Reduce the SWR in the feed line to the antenna
B. Reduce the power dissipation in the feedline to the antenna
C. Increase power transfer from the transmitter to the feed line
D. All these choices are correct

**G4A07.** What happens as a receiver’s noise reduction control level is increased?
A. Received signals may become distorted
B. Received frequency may become unstable
C. CW signals may become severely attenuated
D. Received frequency may shift several kHz

**G4A08.** What is the correct adjustment for the LOAD or COUPLING control of a vacuum tube RF power amplifier?
A. Minimum SWR on the antenna
B. Minimum plate current without exceeding maximum allowable grid current
C. Highest plate voltage while minimizing grid current
D. Desired power output without exceeding maximum allowable plate current

**G4A09.** What is the purpose of delaying RF output after activating a transmitter’s keying line to an external amplifier?
A. To prevent key clicks on CW
B. To prevent transient overmodulation
C. To allow time for the amplifier to switch the antenna between the transceiver and the amplifier output
D. To allow time for the amplifier power supply to reach operating level

**G4A10.** What is the function of an electronic keyer?
A. Automatic transmit/receive switching
B. Automatic generation of dots and dashes for CW operation
C. To allow time for switching the antenna from the receiver to the transmitter
D. Computer interface for PSK and RTTY operation

**G4A11.** Why should the ALC system be inactive when transmitting AFSK data signals?
A. ALC will invert the modulation of the AFSK mode
B. The ALC action distorts the signal
C. When using digital modes, too much ALC activity can cause the transmitter to overheat
D. All these choices are correct

**G4A12.** Which of the following is a common use of the dual-VFO feature on a transceiver?
A. To allow transmitting on two frequencies at once
B. To permit full duplex operation – that is, transmitting and receiving at the same time
C. To transmit on one frequency and listen on another
D. To improve frequency accuracy by allowing variable frequency output (VFO) operation

**G4A13.** What is the purpose of using a receive attenuator?
A. To prevent receiver overload from strong incoming signals
B. To reduce the transmitter power when driving a linear amplifier
C. To reduce power consumption when operating from batteries
D. To reduce excessive audio level on strong signals

**Answer Key**

| | | | |
|---|---|---|---|
| G4A01: B | G4A02: C | G4A03: C | G4A04: B |
| G4A05: C | G4A06: C | G4A07: A | G4A08: D |
| G4A09: C | G4A10: B | G4A11: B | G4A12: C |
| G4A13: A | | | |